

How is Primate Diet Distinct?

Our origin as primates means that we originally evolved as specialized predators in an arboreal environment. As adaptations to that ecological niche, we have flexible shoulders, dexterous hands, opposable thumbs, keen vision, and well-developed cognitive abilities. These traits set the stage for the behavioral adaptations that humans later developed to implement our specialized food-gathering and -processing strategies.

Primate Digestion

From our primate ancestors, we inherited a single stomach; we are **monogastric**. This means that we are inefficient digesters of cellulose, the primary component of many types of plant tissue. Grasses and some leaves are high in cellulose, and it composes most of a wood molecule. This is why many of these plant foods that other species favor are inedible to humans. Some primate species have evolved the ability to digest cellulose better than humans, however, and can feed on leaves that we cannot. But this is an evolved specialization in those primate lineages and is not ancestral to primates as a clade.

Monogastric animals have a single stomach, in contrast with ruminant animals (such as cows), which have multiple stomach chambers for specialized, multistage fermentation of plant foods, especially cellulose.

Primate Dentition

The dentition, or number and shape of teeth, of primates varies significantly depending on the food specialization of a particular species. These specializations range from a focus on anterior (frontal) dentition, used to bite through soft materials, to a focus on posterior (rear) dentition, with large molars used to grind tough foods. Our basic dental formula is 2:1:2:3, which means that each quadrant of the mouth contains (counting from front to back) two incisors, one canine, two premolars, and three molars. Earliest primates likely had a dental formula of 2:1:4:3, meaning that we have lost two premolars since that time; this occurred prior to the separation of the lineage

that includes humans, chimpanzees, gorillas, orangutans, and gibbons, as they all share our dental formula. New World monkeys (including marmosets, howler, and spider monkeys) still have three premolars.



Anterior (front) dentition of a ring-tailed lemur. The ancestral canines have become much like incisors, and ancestral premolars have become canine-like. This lemur is an omnivore, eating fruits, leaves, insects, and small vertebrates (birds and lizards).

Source: [Wikimedia Commons](https://commons.wikimedia.org/wiki/File:Lemur_catta_toothcomb.jpg)

(https://commons.wikimedia.org/wiki/File:Lemur_catta_toothcomb.jpg)



Dentition of a *Hamadryas* baboon. Note the human-like arrangement of teeth but exaggerated canines, used in threat displays (such as this one) and competition for mates. This species lives in the arid savannas of northeastern Africa and eats an omnivorous diet, including a variety of plant parts (flowers, seeds, grasses, roots, bark, and leaves); invertebrates (insects and worms); and smaller vertebrates (birds, reptiles, and small-bodied mammals).

Source: [University of California, Berkeley](https://news.berkeley.edu/2016/07/11/changes-in-primate-teeth-linked-to-rise-of-monkeys/).

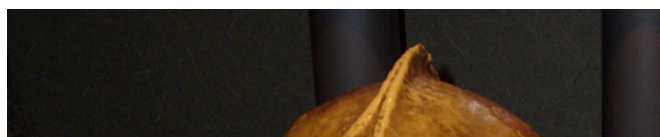
(<https://news.berkeley.edu/2016/07/11/changes-in-primate-teeth-linked-to-rise-of-monkeys/>)



A cast of modern human upper dentition. Note that only two molars are present, consistent with the current evolutionary state of our species, in which small mouths have crowded the dentition to the point where third molars (wisdom teeth) often require surgical removal. Humans have a generalist dentition suitable for eating a broad, omnivorous diet that requires less mastication (chewing in the mouth) than a diet heavy in cellulose might need.

Source: [Wikimedia Commons](https://commons.wikimedia.org/wiki/File:Upper_Jaw_Dentition.jpg).

(https://commons.wikimedia.org/wiki/File:Upper_Jaw_Dentition.jpg)





A cast of a fossil skull and mandible of *Paranthropus boisei*, a “robust australopithecine” that was a sister lineage to that of modern humans. Note the very large, molar-like premolars. This species is inferred to have eaten a diet rich in low-quality, abrasive vegetable foods, including grasses and underground storage organs (roots and tubers).

Source: Wikimedia Commons